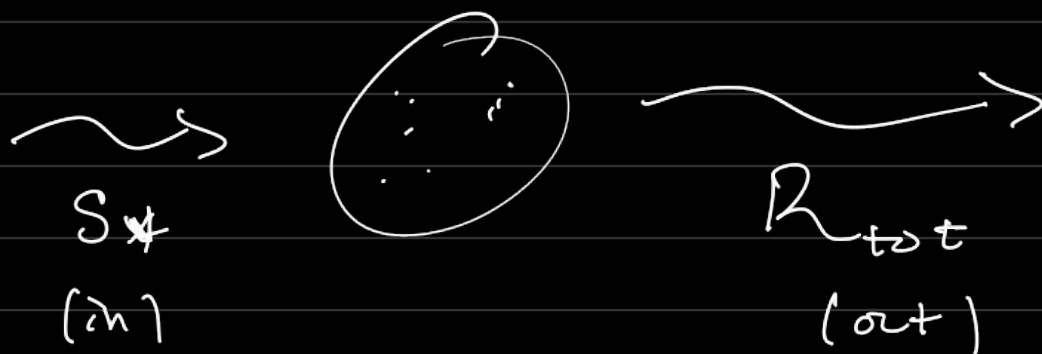


* Recall the Strömgren radius is defined as the radius where we have equilibrium in ionization & recombination



$$S^* = R_{\text{tot}}$$

* In the lecture we looked at the case for constant n_H . Here n_H varies. This affects R_{tot} . Let's see how

Recall for a single recombination

$$R = \alpha_{\text{rec}} n_H \quad \leftarrow n_p = n_e = n_H$$

• look at some ΔR



• Inside ΔR there are N_e electrons which will combine so the total recomb. in this small volume

$$\Delta R_{\text{tot}} = N_e \cdot R = N_e \alpha n_H$$

$$N_e = \underset{\uparrow}{n_e} \Delta V = \underset{\uparrow}{n_H} (4\pi R^2 \Delta R)$$

$$n_e = n_H!$$

plug this into ΔR_{tot}

$$\begin{aligned}\Delta R_{tot} &= n_H (4\pi R^2 \Delta R) \propto n_H \\ &= \alpha n_H^2 4\pi R^2 \Delta R\end{aligned}$$

Now, the total is given by the integral

$$R_{tot} = \int \Delta R_{tot} = \int_{R_*}^{R_s} \alpha n_H^2 4\pi R^2 dR$$

Since $\Delta R \rightarrow dR$.

$$R_{tot} = 4\pi\alpha \int_{R_*}^{R_s} n_H^2 R^2 dR$$

* Note: if n_H is constant then we get the expression from our lesson. However, the question states we have

$$n_H = n_0 \left(\frac{R}{R_0} \right)^{-1/2}$$

so we plug this in

$$\begin{aligned}R_{tot} &= 4\pi\alpha \int_{R_*}^{R_s} \left(n_0 \left(\frac{R}{R_0} \right)^{-1/2} \right)^2 R^2 dR \\ &= 4\pi\alpha n_0^2 \int_{R_*}^{R_s} \left(\frac{R_0}{R} \right) R^2 dR \\ &= 4\pi\alpha n_0^2 R_0 \int_{R_*}^{R_s} R dR \\ &= 2\pi\alpha n_0^2 R_0 (R_s^2 - R_*^2)\end{aligned}$$

Since the problem tells us that

$R_s \gg R_*$ then

$$(R_s^2 - R_*^2) = R_s^2 \left(1 - \frac{R_*^2}{R_s^2}\right)$$

$$\left(\frac{R_*}{R_s}\right)^2 \approx 0$$

$$\text{so } (R_s^2 - R_*^2) \approx R_s^2$$

$$\underline{R_{\text{tot}}} = 2\pi\alpha n_0^2 R_0 R_s^2$$

* Finally, we wanted: $S_* = R_{\text{tot}}$, so

$$S_* = 2\pi\alpha n_0^2 R_0 R_s^2$$

solve for R_s :

$$R_s = \sqrt{\frac{S_*}{2\pi\alpha n_0^2 R_0}}$$

$$(c) \lambda = 4861 \text{ \AA}$$

$$\frac{1}{2} R_{\text{tot}} = \frac{\# \text{ photons}}{s} \quad | \quad R_{\text{tot}} = S_*$$

$$\langle E \rangle = h\nu = \frac{hc}{\lambda}$$

$$\frac{1}{2} S_* \langle E \rangle = \frac{1}{2} S_* \left(\frac{hc}{\lambda}\right) = L$$

$$L = \frac{1}{2} \left(\frac{hc}{\lambda}\right) (S_*)$$

